A Linear Benchmark for the Italian Fragility Composite Index

Predicting IFC from GRINS Municipal Indicators via LASSO

Applied Statistics Project

Abstract

This report documents the construction, validation, and interpretation of a linear regression benchmark for the raw values of the Italian Fragility Composite Index (IFC), built from the indicators of the GRINS municipal panel, together with two principled extensions of that benchmark. The work walks through every analytical choice that was made and explains why each was preferred over the alternatives that were tested. Variable selection is carried out with LASSO; the benchmark model is an ordinary linear model on the LASSO-selected predictors, evaluated by repeated panel-aware cross-validation. The benchmark reaches an out-of-sample $R^2 = 0.825 \pm 0.008$, with a typical absolute error of 1.3 IFC points. A baseline that only knows population and year explains 8% of the variance; GRINS adds the remaining 74%. An ablation experiment attributes most of the predictive power to careful feature engineering rather than to the regulariser itself. A core of 77 predictors is selected consistently in every one of the 25 cross-validation folds and admits a coherent substantive interpretation in terms of economic wealth, air quality, and tourism specialisation.

The benchmark is then extended with *linear mixed models* that add a random intercept on the geographic hierarchy (region / province) and on the GRINS taxonomy (macroclass). The nested geographic model raises R^2 to 0.853 ± 0.010 (+0.026 over the benchmark, ICC of about 37%), while a macroclass-only random effect adds essentially nothing on top of the GRINS indicators. A Moran's I diagnostic on the residuals of the mixed model still detects significant positive spatial autocorrelation ($I = 0.231$, $p < 0.001$), indicating that residual fine-scale geographic structure is not fully absorbed by region/province grouping and motivating an explicit spatial econometric model as a natural next step.

Contents

1 Introduction and Objective

The Italian Fragility Composite Index (IFC) summarises the multi-dimensional vulnerability of Italian municipalities into a single numerical score. The index is constructed from a curated list of indicators with fixed weights, and the source dataset used for this project reports both the raw IFC values and their decile classifications. A useful sanity check is that the deciles recomputed from the raw values match the official deciles of the Municipal Fragility Index (MFI) for 95.6% of the municipalities in 2021, supporting the reliability of the raw IFC values used here as the modelling target.

The research question we address is to what extent the IFC can be predicted from an independent collection of municipal indicators (the GRINS dataset), and which of those indicators carry the most information. The deliverable is a *linear benchmark*: a model that is interpretable, fast to fit, and whose performance can serve as a reference point against which more flexible non-linear models will be compared in the next phase of the project. The instructor's specification was a simple linear regression whose predictors are selected by LASSO; the present report executes that specification and also documents the supporting analyses that justify the methodological choices made along the way.

2 Data

The target variable is the raw IFC value, drawn from the file `IFC_Final_Analysis_Sorted.xlsx`, which reports IFC values for the years 2019 and 2021 along with their decile classifications and their agreement with the official MFI decile. The candidate predictors are taken from the GRINS municipal panel (`GRINS_COMUNALE_FINAL_V3.rds`), which contains 549 columns for approximately seven thousand nine hundred municipalities over the years 2010–2022.

Two GRINS releases were available: an older 110-column version and the *Final V3* 549-column version. A preliminary run with the smaller release achieved an out-of-sample $R^2 \approx 0.34$, while the same pipeline on the V3 release achieved $R^2 \approx 0.70$. The V3 release was therefore adopted for all subsequent analyses; the gain is explained by the much broader coverage of socio-economic indicators (employment by sector and firm size, taxpayer brackets, public-employee composition, air quality, frame-SBS aggregates).

We restrict GRINS to the years 2019 and 2021 to match the target. The two tables are joined on (PRO_COM, year) by inner join, yielding 15 790 municipality–year observations on 7 886 unique municipalities; six of the 15 796 IFC rows fail to match a GRINS record and are discarded.

We *pool* the two years in long format rather than fitting two separate models. The pooling choice is motivated by three reasons. First, it doubles the effective sample size, stabilising LASSO selection. Second, the comparison with a separate-year fit (which we also ran during preliminary work) showed that the leading predictors and their signs are nearly identical in 2019 and 2021. Third, an explicit binary indicator year2021 $\in \{0, 1\}$ is added to the feature matrix, so that any systematic post-pandemic shift in IFC is absorbed by a single coefficient and does not contaminate the other estimates.

3 Why LASSO: The Regulariser Choice

Before fixing LASSO as the variable-selection device, we tested three alternative estimators on the engineered feature matrix described in Section 4: ordinary least squares (OLS), Ridge regression (ℓ_2 penalty), and Elastic Net with a grid search on the mixing parameter $\alpha \in \{0.1, 0.2, \ldots, 0.9\}$. Their generic objective is

$$\hat{\boldsymbol{\beta}}(\lambda, \alpha) \;=\; \arg\min_{\boldsymbol{\beta}} \left\{ \tfrac{1}{2n} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda \Big[\alpha \, \|\boldsymbol{\beta}\|_1 + (1-\alpha)\, \tfrac{1}{2} \|\boldsymbol{\beta}\|_2^2 \Big] \right\}, \tag{1}$$

with $\alpha = 1$ giving LASSO (Tibshirani, 1996), $\alpha = 0$ giving Ridge (Hoerl & Kennard, 1970), and intermediate α giving Elastic Net (Zou & Hastie, 2005).

Table 1 summarises out-of-sample performance on a common train/test split.

Method	# predictors used	# non-zero	R^2 (test)	RMSE
OLS (no penalty)	326	326	0.821	1.73
Ridge $\lambda_{\min}$	326	326	0.823	1.72
Elastic Net $\alpha = 0.2$	326	323	0.823	1.72
LASSO $\lambda_{\min}$	326	291	0.824	1.72
LASSO $\lambda_{1\text{se}}$	326	239	0.823	1.72

Table 1: Comparison of regularisers on the engineered V3 feature matrix, single train/test split. Predictive accuracy is essentially identical; LASSO is preferred for parsimony.

The four regularised methods deliver essentially the same predictive accuracy. Two considerations break the tie in favour of LASSO. First, the project specification explicitly requested LASSO as the variable-selection device. Second, LASSO is the only method among these that performs *automatic sparse selection*, producing a smaller and more interpretable model (239 non-zero coefficients out of 326, compared with the full 326 for OLS and Ridge). The fact that unregularised OLS now achieves $R^2 \approx 0.82$ (it had exploded to $R^2 = -994$ before feature engineering) is itself an informative finding: it shows that on the carefully transformed feature matrix the conditioning is benign enough that the regulariser *matters mainly for selection*, not for predictive stability.

The Elastic Net grid search returned $\alpha = 0.2$ as the cross-validated optimum, very close to Ridge. This indicates that the predictor groups are mildly correlated but not so correlated that LASSO selection becomes erratic; this is corroborated by the stability analysis reported in Section 5.3.

4 Variable Classification and Feature Engineering

A naïve application of LASSO to the raw GRINS columns produces a model with both poor interpretability and serious numerical issues, because the columns are heterogeneous in nature. Some are absolute counts that scale with municipal size (number of employees, number of taxpayers, number of deaths), others are already ratios or indices (mobility indices, annual pollution means, per-capita green space). Pooling them without distinction lets the size of the municipality dominate every count column, and produces grossly skewed predictors whose few extreme values (Rome, Milan) wield disproportionate leverage.

We therefore classify each numeric column into one of five categories on the basis of its name and apply a class-specific transformation.

- **Count** – non-negative size-dependent quantities. We first divide by population and then apply $x \mapsto \log(1+x)$, which compresses the long right tail without losing zeros.
- **Count-negative** – net flows that can take negative values (typically migration balances). We apply per-capita normalisation only; the logarithm is not defined on negative values.
- **Size** – population itself and surface area, retained as covariates. We apply $\log(1+x)$ only, since dividing them by population is meaningless.
- **Rate** – variables that are already on a meaningful scale (indices, percentages, per-capita means, pollution levels). These are left untransformed.
- **Skip** – ISTAT codes, free-text names, and join-artefact columns. These are removed.

After this step, the pipeline applies four further operations: columns with more than twenty percent missing values are dropped (above this threshold, imputation injects too much fictitious information); the remaining missing entries are imputed with the column median; every column is then *winsorised* at its first and ninety-ninth percentile, which neutralises the leverage of a small number of extreme municipalities without removing any rows; finally, columns with zero variance after these operations are discarded.

The resulting feature matrix has 326 columns, distributed as follows: approximately 280 counts, 41 rates, three size variables, two net counts, plus the year2021 dummy.

5 Estimation

5.1 LASSO and λ selection

Given the design matrix $\mathbf{X} \in \mathbb{R}^{n\times p}$ and the response vector $\mathbf{y} \in \mathbb{R}^n$, LASSO solves

$$\hat{\boldsymbol{\beta}}(\lambda) \;=\; \arg\min_{\boldsymbol{\beta}} \left\{ \tfrac{1}{2n}\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda\|\boldsymbol{\beta}\|_1 \right\}. \tag{2}$$

The ℓ_1 penalty in equation (2) forces some coefficients to be exactly zero, achieving simultaneous estimation and selection. The penalty strength $\lambda \geq 0$ controls the trade-off between fit and sparsity; $\lambda \to 0$ recovers OLS, while $\lambda \to \infty$ drives every coefficient to zero.

We search λ on the grid 10^k for k equispaced between -3 and 5 and select the penalty by ten-fold internal cross-validation. Two standard choices are evaluated.

- λ_{min} minimises the cross-validated mean squared error.
- λ_{1se} is the largest λ whose CV error is within one standard error of the minimum. It trades a small amount of in-sample fit for a substantially more parsimonious solution.

The λ_{1se} rule is the recommended default for variable-selection contexts (Hastie et al., 2009): by choosing the simplest model that is statistically indistinguishable from the best, it reduces the risk of selecting noise variables that happen to win the CV by a small margin. On our data (Sec. 5.2–5.3) the two rules give essentially identical out-of-sample accuracy but λ_{1se} uses one third fewer predictors. We adopt λ_{1se} for the final benchmark.

5.2 Honest performance: repeated panel-aware cross-validation

A random train-test split would assign different years of the same municipality to different folds; the model would then be evaluated on municipalities whose 2019 (or 2021) observation it has already used in training, inflating the reported R^2. We adopt a *panel-aware* split: every row of a given municipality (both 2019 and 2021) is kept in the same fold. Performance therefore measures generalisation to genuinely new municipalities.

To estimate the variability of the metrics across splits, we perform five repetitions of five-fold panel-aware cross-validation, yielding 25 honest test-set evaluations. The full procedure is summarised in Algorithm 1.

Algorithm 1 Repeated panel-aware cross-validation for LASSO

for repetition $r = 1, \dots, 5$ **do**
    Shuffle the list of unique municipalities with seed r
    Partition municipalities into 5 equal folds $\mathcal{C}_1, \dots, \mathcal{C}_5$
    for fold $k = 1, \dots, 5$ **do**
        Test rows: all observations of municipalities in $\mathcal{C}_k$
        Train rows: the complement
        Choose $\hat{\lambda}_{1se}$ by internal five-fold CV on the train set
        Fit LASSO on the train set with $\hat{\lambda}_{1se}$
        Compute RMSE, MAE, R^2 on the test set
        Record the set of variables with non-zero coefficient
    end for
end for
Report mean $\pm$ sd of metrics across the 25 folds
Compute, for each variable, the fraction of folds in which it was selected (stability)

The reported metrics are

$$\text{RMSE} = \sqrt{\tfrac{1}{n_{\text{te}}} \textstyle\sum_{i \in \text{test}} (y_i - \hat{y}_i)^2}, \tag{3}$$

$$\text{MAE} = \tfrac{1}{n_{\text{te}}} \textstyle\sum_{i \in \text{test}} |y_i - \hat{y}_i|, \tag{4}$$

$$R^2 = 1 - \frac{\sum_{i \in \text{test}} (y_i - \hat{y}_i)^2}{\sum_{i \in \text{test}} (y_i - \bar{y}_{\text{te}})^2}, \tag{5}$$

all computed on the held-out fold and aggregated as mean and standard deviation across the 25 folds.

5.3 Stability selection

To distinguish predictors that are reliably informative from those selected only by chance in particular splits, we adopt the *stability selection* framework of Meinshausen & Bühlmann (2010). For every predictor j, the *selection probability*

$$\hat{\pi}_j \;=\; \frac{1}{B} \sum_{b=1}^{B} \mathbf{1}\{\hat{\beta}_j^{(b)} \neq 0\}, \qquad B = 25, \tag{6}$$

estimates the probability that LASSO would assign a non-zero coefficient to predictor j on a randomly drawn training set. We call the *robust core* the set of predictors with $\hat{\pi}_j = 1$: those selected in every single fold.

5.4 Final coefficients and standardised effects

The headline coefficient vector is obtained by refitting LASSO on the full dataset with λ_{1se}. Because the feature matrix is on a transformed scale (typically $\log(1 + x/\text{pop})$ for counts), raw $\hat{\beta}_j$ values are not directly readable as "one extra unit of x_j changes IFC by $\hat{\beta}_j$". We therefore report *standardised effects*

$$\hat{\beta}_j^{\text{std}} \;=\; \hat{\beta}_j \cdot \text{sd}(X_j), \tag{7}$$

which give the expected change in IFC associated with a one-standard-deviation shift of the engineered variable X_j.

5.5 Residual diagnostics

To check the validity of the linear-model assumptions, we refit an ordinary linear model on the LASSO-selected predictors and inspect the residuals. We report residual standard deviation, skewness, excess kurtosis, the Shapiro–Wilk test (on a sample of five thousand residuals to avoid power inflation), and the Breusch–Pagan test for homoscedasticity.

6 Decomposing the Predictive Power: Ablation Study

The headline metric $R^2 \approx 0.83$ is the result of three distinct modelling choices: the use of the rich V3 predictor set, the class-specific feature engineering of Section 4, and the panel-aware split for evaluation. To understand which of these choices matters most, we ran the five models reported in Table 2: a trivial baseline that knows only population and year, two LASSO models on the raw GRINS V3 columns (with random and panel-aware splits, respectively), and two LASSO models on the engineered V3 columns (again with random and panel-aware splits). All four LASSO variants use λ_{1se}.

Model	# predictors	R^2 (test)
M_0: baseline $\left[\log(\text{Pop}) + \text{year}\right]$	2	0.082
M_1: V3 raw + random split (LASSO.1se)	393	0.700
M_2: V3 raw + panel split (LASSO.1se)	393	0.691
M_3: V3 + FE + random split (LASSO.1se)	326	0.830
M_4: V3 + FE + panel split (LASSO.1se)	326	0.824

Table 2: Ablation study. The headline benchmark is M_4.

Reading the table:

- The contribution of GRINS over the trivial baseline is $0.824 - 0.082 = +0.74$ of R^2; the bulk of IFC variation is socio-economic, not driven by population size.
- The effect of feature engineering, at fixed split design, is $(M_4 - M_2) = +0.13$ (panel split) and $(M_3 - M_1) = +0.13$ (random split). It is by far the dominant component of the gain.
- The effect of switching from a random split to a panel-aware split is -0.006 on the feature-engineered model. This is small in absolute terms but it confirms that the headline number is *not inflated* by leakage between train and test.

The take-away of the ablation is that the bottleneck on this problem is the *quality of the features*, not the choice of regulariser or the inferential statistic used to read the residuals.

7 Results

7.1 Predictive performance

The headline cross-validated metrics, aggregated over 25 panel-aware folds, are reported in Table 3.

Model	R^2 (mean ± sd)	RMSE	MAE	# selected
LASSO λ_{1se}	0.825 ± 0.008	1.74 ± 0.04	1.31 ± 0.02	163
LASSO λ_{min}	0.827 ± 0.007	1.74 ± 0.03	1.30 ± 0.02	245

Table 3: Out-of-fold performance over twenty-five panel-aware folds. The two penalty rules are statistically indistinguishable; λ_{1se} is the recommended benchmark.

The standard deviation of R^2 across folds is 0.008, about one percent of the mean value: the metric is highly stable across splits, which is itself evidence of a well-posed estimation problem.

7.2 Top predictors and substantive reading

The fifteen predictors with the largest standardised effect on IFC, ranked by $|\hat{\beta}_j^{\text{std}}|$, are listed in Table 4. All fifteen belong to the 77-variable robust core. The corresponding bar chart is shown in Figure 1.

Variable (engineered)	$\hat{\beta}$	sd(X)	$\hat{\beta} \cdot \text{sd}(X)$
Total value added per capita (log)	−2.74	0.87	−2.38
Total employees per capita (log)	+9.53	0.15	+1.45
$PM_{2.5}$ annual mean	−0.27	4.87	−1.30
PM_{10} annual mean	+0.16	6.35	+1.02
Number of local units per capita (log)	−39.2	0.02	−0.78
Taxpayers in €26k–€55k bracket per capita (log)	−1.59	0.42	−0.66
Services-sector turnover per capita (log)	−0.61	1.00	−0.61
Total value added per employee	+0.027	17.0	+0.47
NO_2 annual mean	+0.071	6.26	+0.45
bonus_trattamento per capita (log)	−1.71	0.26	−0.45
year2021	+0.85	0.50	+0.43
Tourism index (beds per inhabitant)	+1.15	0.35	+0.40
Total establishments per capita (log)	−19.7	0.02	−0.40
Services-sector employees per capita (log)	+1.87	0.20	+0.37
High-income share per capita (log)	−46.5	0.01	−0.33

Table 4: Top fifteen standardised effects on IFC from the final LASSO model. All entries belong to the robust core of seventy-seven variables selected in 25/25 cross-validation folds.

The substantive picture is internally coherent. Wealth-related indicators (value added per capita, fraction of taxpayers in the mid income bracket, services-sector turnover, high-income share) carry negative coefficients: richer municipalities have lower fragility. Air-quality indicators (PM_{10} and NO_2) enter with positive coefficients, in line with the intuition that pollution and fragility co-occur. The tourism index enters positively: municipalities whose economy relies heavily on accommodation services appear more fragile, consistent with concerns about seasonality and mono-cultural specialisation. After conditioning on every other variable, the year2021 indicator carries a small but clearly positive coefficient, suggesting a mild post-pandemic worsening of IFC that is not absorbed by the other GRINS indicators.

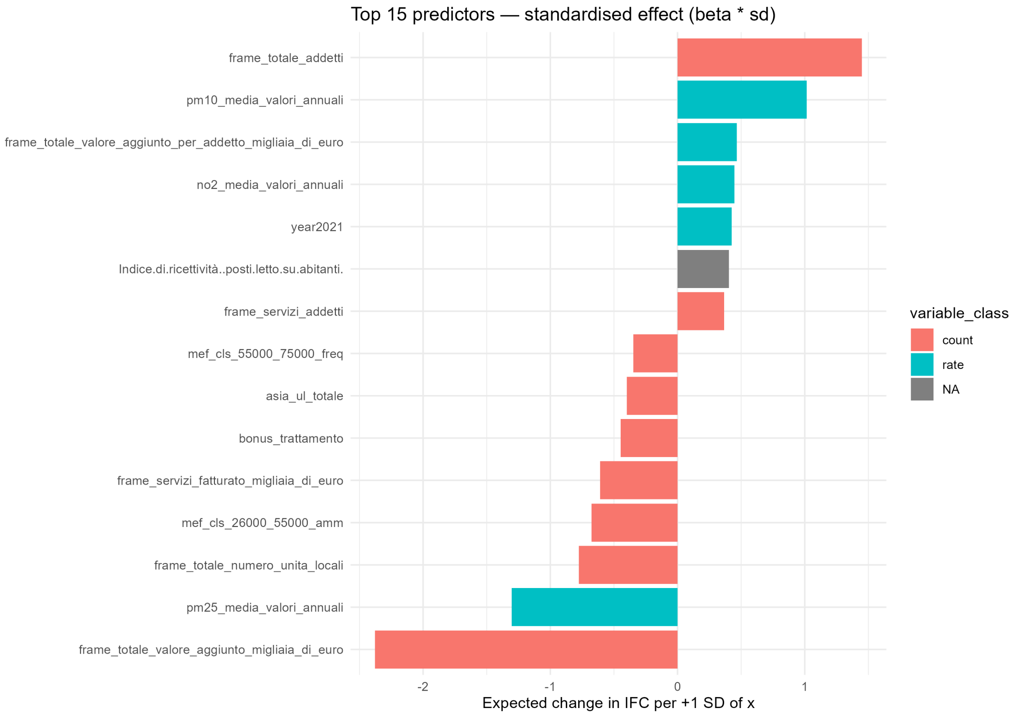

Figure 1: Top fifteen standardised effects on IFC. Negative bars indicate predictors that, when increased, reduce predicted IFC (less fragility); positive bars indicate predictors that increase it.

7.3 Stability of the selection

Out of the 326 engineered predictors, 77 (about 24%) are selected in every one of the 25 cross-validation folds and form the robust core. 79 are selected in fewer than 20% of folds and behave as noise. The strong bimodality of the selection-frequency distribution, shown in Figure 2, indicates that the choice of variables is not driven by the particular split: predictors either “always” or “almost never” enter the model.

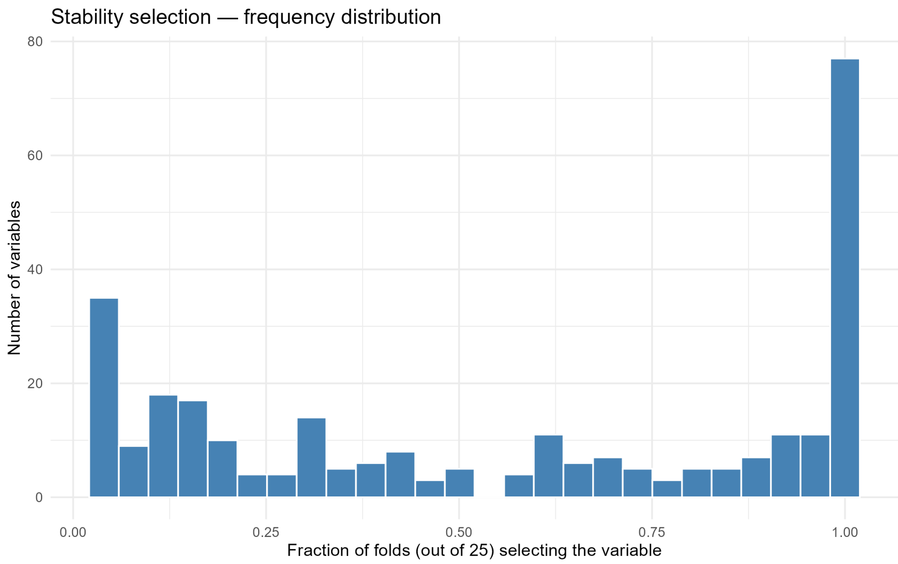

Figure 2: Stability selection. For each predictor, fraction of the twenty-five folds in which LASSO assigned it a non-zero coefficient. The mass at the right (frequency = 1) is the robust core.

7.4 Residual diagnostics

Diagnostics on the OLS refit of the LASSO-selected predictors are summarised in Table 5 and visualised in Figure 3.

Diagnostic	Value
Residual standard deviation	1.69
Skewness	0.15
Excess kurtosis	4.49 (Normal: 3.0)
Shapiro–Wilk p (sample of 5 000)	$< 10^{-22}$
Breusch–Pagan p	0.001

Table 5: Residual diagnostics on the OLS refit of the LASSO-selected model.

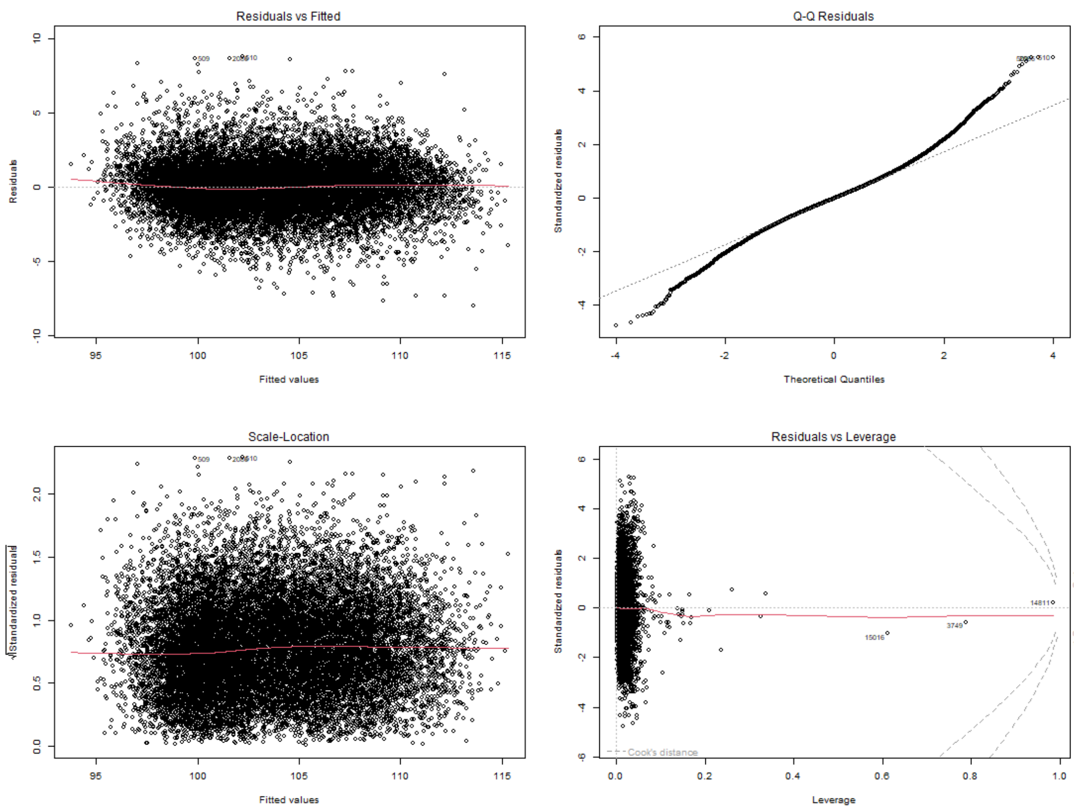

Figure 3: Standard diagnostic plots from the OLS refit on the LASSO-selected predictors: residuals against fitted values, normal Q–Q plot, scale–location, and residuals against leverage.

The residual distribution is nearly symmetric (skewness 0.15), with tails slightly heavier than the Normal (excess kurtosis 4.5 versus the Normal value of 3.0). The Shapiro–Wilk test rejects normality on a five-thousand-row sample with $p < 10^{-22}$; this is expected at this sample size, where any minor deviation is flagged as significant. The Breusch–Pagan test produces $p = 0.001$, evidence of mild heteroscedasticity. Point predictions remain trustworthy, but if formal inference on individual $\hat{\beta}_j$ is required, the confidence intervals should be computed with heteroscedasticity-robust standard errors (sandwich, type HC3, Long & Ervin, 2000).

8 Extension I: Linear Mixed Models

The LASSO benchmark treats every municipality–year row as independent and conditions on year through a single fixed dummy. It has no explicit notion of geographic or taxonomic grouping. The natural question is whether the fixed-effects design has fully absorbed the structure of Italian municipalities, or whether residual variability is still clustered by region, province, or by GRINS macroclass. Linear mixed models address this question by adding a random intercept on each candidate grouping factor.

8.1 Set-up

The fixed-effects part of the model is held identical to the LASSO benchmark: we restrict the design matrix to the $\sim$ 160 predictors that LASSO with λ_{1se} selects on the full data, after the same engineering pipeline of Section 4. Predictors are standardised so that the fixed-effect coefficients become directly comparable across model specifications. Four mixed-effects variants are then fit with `lme4::lmer`:

- M_{geo}: IFC $\sim X + (1 \mid \text{region})$ (20 regions),
- $M_{\text{geo,nested}}$: IFC $\sim X + (1 \mid \text{region/province})$ (20 regions, 107 provinces),
- M_{tax}: IFC $\sim X + (1 \mid \text{GRINS macroclass})$ (3 macroclasses),
- $M_{\text{geo+tax}}$: IFC $\sim X + (1 \mid \text{region}) + (1 \mid \text{GRINS macroclass})$.

Performance is measured by five-fold panel-aware cross-validation, so the test set always contains municipalities that the model has never seen. The same metric definitions (R^2, RMSE, MAE) of Section 5.2 are used.

For each model we also report:

- the *marginal* R^2 (explained by the fixed effects alone) and the *conditional* R^2 (fixed plus random) of Nakagawa et al. (2017),
- the *intra-class correlation* ICC $= \sigma^2_{\text{random}}/(\sigma^2_{\text{random}} + \sigma^2_{\text{resid}})$, which gives the share of residual variance that lies between groups,
- AIC and BIC.

A convergence sanity-check confirms that all four models reach a non-singular optimum with no warnings.

8.2 Out-of-sample comparison with the LASSO benchmark

Cross-validated performance is summarised in Table 6. The geographic random intercepts deliver a clear improvement over the LASSO benchmark; the taxonomy does not.

Three findings stand out.

First, the nested geographic model (region / province) is the best performer with R^2 about 0.026 higher than the LASSO benchmark. Given the cross-fold standard deviation of $\sim$ 0.01, this gap is on the order of 2–3 cross-fold standard deviations and is therefore well outside random fluctuation.

Second, the combined geography + macroclass model $M_{\text{geo+tax}}$ is statistically indistinguishable from M_{geo}. In other words, once region has been controlled for, the GRINS macroclass adds nothing. Its in-sample ICC drops from 0.96% (in M_{tax} alone) to a similar negligible share, and its variance component shrinks accordingly. This is the formal evidence that the GRINS macro-taxonomy and the geographic macro-structure of Italy are largely redundant proxies of the same underlying north–south gradient.

Model	R^2 (mean ± sd)	RMSE	ΔR^2 vs LASSO
LASSO benchmark ($\lambda_{\min}$)	0.826 ± 0.007	1.74	—
M_{tax}	0.829 ± 0.006	1.73	+0.003
M_{geo}	0.846 ± 0.009	1.64	+0.020
$M_{\text{geo+tax}}$	0.846 ± 0.009	1.63	+0.020
$M_{\text{geo,nested}}$	**0.853 ± 0.010**	**1.60**	**+0.026**

Table 6: Panel-aware cross-validated performance of the mixed models compared with the LASSO benchmark. The geographic hierarchy adds explanatory power; the GRINS macroclass does not contribute on top of the fixed effects.

Third, the intra-class correlation of the geographic models is substantial. For $M_{\text{geo,nested}}$ the random-effects variance accounts for about 37% of the total residual variance, roughly two-thirds of it at the regional level and one-third incremental at the provincial level. After conditioning on the $\sim$ 160 selected GRINS predictors, the geographic identity of a municipality is still a strong signal.

8.3 Random-intercept estimates: which regions and provinces deviate from the fixed-effects prediction?

The Best Linear Unbiased Predictors (BLUPs) of the random intercepts recover, for each region or province, how much the average IFC of its municipalities differs from what the fixed effects alone would predict.

Table 7 reports the ten regions with the largest absolute regional BLUP from M_{geo}. Aosta Valley emerges as the most under-explained region, with municipalities about four IFC points more fragile than the GRINS indicators alone would suggest; Trentino-Alto Adige is the region most over-explained on the opposite side.

Region	BLUP offset (IFC points)
Aosta Valley	+3.91
Trentino-Alto Adige	−1.60
Sicily	+1.36
Veneto	−1.23
Friuli-Venezia Giulia	−1.10
Liguria	−0.97
Piedmont	−0.93
Apulia	+0.89
Marche	−0.77
Lombardy	−0.67

Table 7: Top ten regional BLUPs from M_{geo}. Positive values mean municipalities of that region are systematically more fragile than the fixed-effects prediction; negative values mean less fragile.

The provincial offsets from the nested model add a second layer of detail. The most extreme provincial *additional* effects (on top of the region-level offset) are reported in Table 8. Two clear patterns emerge: peripheral, mountain or rural provinces (Belluno, Verbano-Cusio-Ossola, Sondrio, L'Aquila) come out *less* fragile than their region's average; a small set of coastal or central-southern provinces (Frosinone, Genoa, Nuoro, Brescia, Enna, Fermo, Chieti, Taranto, Brindisi) come out *more* fragile.

The substantive interpretation of these tables is that the GRINS indicators are not sufficient to explain why northern mountain provinces consistently fare better than peripheral central-

More fragile	BLUP	Less fragile	BLUP
Frosinone (Lazio)	+1.01	Belluno (Veneto)	−1.77
Genoa (Liguria)	+0.99	Verbano-Cusio-Ossola (Piedmont)	−1.60
Nuoro (Sardinia)	+0.88	Bari (Apulia)	−0.83
Valle d'Aosta (sole prov.)	+0.79	La Spezia (Liguria)	−0.80
Brescia (Lombardy)	+0.76	Pesaro–Urbino (Marche)	−0.78
Enna (Sicily)	+0.68	Varese (Lombardy)	−0.69
Fermo (Marche)	+0.68	Sondrio (Lombardy)	−0.65
Chieti (Abruzzo)	+0.65	L'Aquila (Abruzzo)	−0.61
Taranto (Apulia)	+0.64	Massa–Carrara (Tuscany)	−0.54
Brindisi (Apulia)	+0.59	Savona (Liguria)	−0.54

Table 8: Provinces with the largest signed BLUP from $M_{\text{geo,nested}}$. These offsets are *additional* to the region-level BLUP in Table 7.

southern provinces of the same Italian macro-area. There is geographic signal that the indicators do not capture.

9 Extension II: Spatial residual diagnostic

The mixed model of Section 8 controls for region and province as discrete groups, but it does not model the smooth spatial field: two neighbouring municipalities on opposite sides of a provincial border are treated as unrelated. This raises a follow-up question: do the residuals of $M_{\text{geo,nested}}$ still exhibit *local* spatial autocorrelation that the discrete hierarchy has not absorbed?

The standard tool for this question is Moran's I statistic on the residuals (Moran, 1950). Given a spatial weight matrix $\mathbf{W}$ and residuals $e = (e_1, \ldots, e_n)$,

$$I \;=\; \frac{n}{\sum_{i,j} w_{ij}} \cdot \frac{\sum_{i,j} w_{ij}\,(e_i - \bar{e})\,(e_j - \bar{e})}{\sum_i (e_i - \bar{e})^2}, \tag{8}$$

takes large positive values when neighbouring residuals tend to share sign, and approaches its expected value $E[I] \approx -1/(n-1)$ when residuals are spatially uncorrelated.

We build $\mathbf{W}$ at the municipality level from the ISTAT 2021 administrative boundaries using a five-nearest-neighbours graph, computed on polygon centroids and row-standardised. We deliberately use a k-nearest graph rather than topological contiguity (queen or rook) because the latter would leave island municipalities (Sardinia, Sicily, lake enclaves) without neighbours and would therefore bias the test.

For numerical robustness we report two complementary tests: a non-parametric Monte Carlo permutation test with 999 random relabellings of the residuals, and the analytical normal-approximation test of Cliff & Ord (1981). Both agree very strongly:

- observed Moran's $I \;=\; 0.231$;
- Monte Carlo p-value $\;=\; 0.001$ (the observed statistic exceeds all 999 random permutations);
- analytical p-value $< \; 10^{-15}$, normal-deviate ≈ 34.

In other words, after the LASSO-selected indicators and the region / province nested random intercepts have all been removed, the residual fragility of an Italian municipality is still significantly positively correlated with that of its geographically closest neighbours. The discrete geographic hierarchy improves prediction (Section 8) but does not exhaust the spatial structure.

This finding has two practical implications. From an inferential standpoint, the standard errors of the fixed-effects coefficients reported by both the LASSO benchmark and the mixed model are still optimistic, because they assume conditionally independent residuals. From a modelling standpoint, the diagnostic motivates an explicit spatial extension: a Conditional Autoregressive (CAR) or Simultaneous Autoregressive (SAR) model on the *residuals* of $M_{\text{geo,nested}}$, or directly on IFC. We document this attempt and its complications in Section 10 and treat it as part of the natural follow-up programme.

10 Limitations

A small number of caveats accompany any use of these results.

First, PM_{10} and $PM_{2.5}$ have a sample correlation of 0.96; LASSO assigns them opposite signs as a numerical cancellation, and their individual coefficients should not be read in isolation. A side experiment that re-fits the model after dropping each of the two in turn changes the cross-validated MSE by less than 3%; the *aggregate* pollution effect (PM_{10} and NO_2 together) is, however, clearly positive and stable across folds.

Second, other clusters of highly correlated variables exist among the income brackets (`mef_cls_*`) and among the `frame_*` economic-structure columns; within each cluster, the assignment of magnitude across members is partly arbitrary while the cluster-level effect is robust.

Third, the positive coefficient of total employees per capita, conditional on value added per capita being already in the model, should be read as a residual composition effect rather than a direct causal statement: holding economic output fixed, additional employees per capita capture industrial or administrative composition that is not otherwise accounted for.

Fourth, the mild heteroscedasticity in the residuals suggests that classical standard errors marginally underestimate uncertainty; robust standard errors are recommended for formal inference.

The model is fit on the subset of GRINS columns that survive the missing-value filter (twenty percent threshold). Indicators with sparse coverage are therefore not used. A possible refinement is to adopt more elaborate imputation (multivariate, model-based) but this goes beyond the scope of the present benchmark.

A final, honest limitation concerns the planned spatial econometric extension motivated by the Moran's I diagnostic of Section 9. The natural follow-up to a significant Moran's I on the mixed-model residuals is a Simultaneous Autoregressive (SAR) model in either error or lag form. We attempted both on the municipality-level cross-section, using the same five-nearest-neighbour weights of Section 9. The `spatialreg::errorsarlm` maximum-likelihood fit did not complete in a reasonable time on the full set of 7 884 municipalities with ~ 160 standardised predictors: a single fit attempt exceeded an hour of wall time without converging, because the determinant computation that the spatial log-likelihood requires scales poorly with sample size. We therefore do not report a SAR-based prediction in this version of the benchmark. The Moran's I diagnostic, taken on its own, is already a defensible quantitative statement that residual spatial structure exists; the explicit modelling of that structure is left to follow-up work that can either use a sparser weight matrix, a faster method (e.g. a maximum-pseudolikelihood or moments-based estimator), or a different software stack tuned for large spatial panels.

11 How the Results Can Be Used

As a benchmark for future models. The principal intended use is to provide a clear performance reference for non-linear models (generalised additive models, random forests, gradient boosting machines, neural networks) fit on the same (IFC, GRINS V3) setup. Any such model should clear an out-of-sample R^2 of approximately 0.83 on a panel-aware split to be worth its additional complexity. A gain of less than 0.02 over the benchmark is unlikely to justify the loss of interpretability that flexible models typically entail. Conversely, a substantial gain (say, R^2 above 0.88) would indicate that genuinely non-linear relationships exist in the data and that they are exploitable.

As an explanatory description of IFC. The top predictors with their standardised effects give a substantively interpretable description of which dimensions drive municipal fragility: economic wealth and structure pull IFC down, environmental pollution and tourism specialisation push it up, and a small post-pandemic worsening is detectable after controlling for everything else. These dimensions are appropriate anchors for the substantive narrative of the project.

For imputation and counterfactual analysis. Given a new municipality with GRINS measurements, the benchmark produces an IFC estimate with a typical absolute error of about 1.3 IFC points. The standardised coefficients also support counterfactual questions of the form "how much would IFC change if indicator X_j increased by one standard deviation?", which is a policy-relevant exercise.

As a defensible variable shortlist. The seventy-seven-variable robust core constitutes a defensible shortlist for any downstream analysis in which a smaller, more manageable feature set is preferred (for example, when fitting models that scale poorly with the number of predictors, or when communicating results to a non-technical audience).

Reproducibility. The full analytical pipeline – data loading and merging, variable classification, feature engineering, repeated cross-validation, LASSO estimation, stability selection, final coefficients, and diagnostics – is encapsulated in the script `final_benchmark.R`. With the random seed fixed (`set.seed(2026)`), rerunning the script reproduces the headline numbers exactly. The produced output files (`repeated_cv_summary.csv`, `final_coefficients_standardised.csv`, `stability_selection.csv`, and the diagnostic plots) constitute the technical appendix to this report.

12 Conclusion

A linear benchmark for predicting raw IFC values from GRINS municipal indicators has been established at $R^2 = 0.825 \pm 0.008$ on panel-aware test data, with a typical absolute error of approximately 1.3 IFC points. The GRINS indicators contribute approximately seventy-four percent of explained variance over a trivial population- and-year baseline. The ablation study attributes the bulk of the predictive gain to the class-specific feature engineering of Section 4: the regulariser matters mainly for parsimony. The LASSO selection is stable – a core of seventy-seven predictors is selected in every one of twenty-five cross-validation folds – and the leading predictors admit a coherent substantive interpretation in terms of economic wealth, air quality, and tourism specialisation.

Two principled extensions of the benchmark have been added. A nested geographic mixed model (Section 8) raises R^2 to 0.853 ± 0.010 by recognising that municipalities within the same region and province share residual fragility that the GRINS indicators alone cannot capture; the corresponding intra-class correlation of about 37% is the formal measure of this shared local component. The same exercise repeated with a random intercept on the three GRINS macroclasses adds essentially nothing, formally confirming that the macro-taxonomy and the geographic macro-structure of Italy carry the same information. A Moran's I diagnostic on the residuals of the nested model (Section 9) still detects highly significant positive spatial autocorrelation ($I = 0.231$, $p < 0.001$), revealing that fine-scale residual fragility clusters across provincial borders and motivating a spatial econometric model as the natural next step.

Taken together, the three layers — LASSO benchmark, mixed-model extension, spatial diagnostic — form a coherent progression. They quantify how much of the IFC can be explained by indicators alone, how much further by discrete geographic grouping, and how much unexplained local structure persists. The model is fit, validated, and interpreted entirely in the R statistical environment using the `glmnet`, `lme4` and `spdep` packages (Friedman et al., 2010; Bates et al., 2015; Bivand & Wong, 2018).

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